

Bioinformatics Bootcamp 2026

Assessment 01

Submitted by: Fatima Rafique

Question 1

A researcher has discovered a DNA sequence that may be associated with a rare genetic disorder. Before conducting laboratory experiments, they want to determine whether this sequence has been reported previously. Which biological database(s) would you recommend for this purpose? Justify your answer by explaining the type of information each database provides and how it would assist the researcher.

When a researcher discovers a DNA sequence that may be linked to a rare genetic disorder, the first step should be to check whether the sequence has already been reported. For this purpose, I would recommend using the NCBI GenBank database along with the EMBL-EBI nucleotide database.

GenBank is one of the largest public repositories of DNA sequences. It contains nucleotide sequences submitted by researchers from around the world, along with information about genes, organisms, and related scientific studies. By searching the newly discovered sequence in GenBank, the researcher can determine whether similar or identical sequences have already been documented.

Another useful resource is the NCBI BLAST tool, which compares a query sequence against millions of sequences stored in databases. BLAST can identify closely related sequences, even if they are not exactly identical. This helps the researcher understand whether the sequence belongs to a known gene or a previously reported genetic region.

If the sequence appears to be associated with a disease, databases such as OMIM (Online Mendelian Inheritance in Man) can also be consulted. OMIM provides detailed information about genetic disorders and their associated genes.

Together, these databases allow the researcher to verify the novelty of the sequence, explore its biological significance, and gather valuable background information before conducting expensive laboratory experiments.

Question 2

A scientist is studying a disease-causing bacterium and wants to compare its genome with those of related bacterial species to identify conserved genes. Which biological database(s) would you recommend for this analysis? Explain why these databases are appropriate and what information they provide.

To compare the genome of a disease-causing bacterium with related bacterial species, I would recommend using the [NCBI Genome database](#) and [Ensembl Bacteria](#).

The [NCBI Genome database](#) contains complete and partially assembled genomes from thousands of bacterial species. It provides access to genomic sequences, gene annotations, and taxonomic information. By obtaining genome data from related species, researchers can perform comparative analyses to identify similarities and differences among organisms.

[Ensembl Bacteria](#) is another valuable resource because it offers genome browsing and comparative genomics tools. It allows researchers to examine gene locations, compare homologous genes across species, and study evolutionary relationships. These features are particularly useful for identifying conserved genes that have remained relatively unchanged over time.

To further investigate conserved regions, the scientist can use [BLAST](#) to compare genes from the target bacterium against genomes of related species. Conserved genes often perform essential biological functions and may play important roles in survival or pathogenicity.

For example, if several related bacteria contain the same antibiotic-resistance gene, it may indicate that the gene is evolutionarily conserved and important for adaptation. By combining information from NCBI Genome, Ensembl Bacteria, and BLAST analyses, the scientist can efficiently identify conserved genes and gain insights into the biology and evolution of the disease-causing bacterium.

Question 3

A biotechnology company has isolated a protein from a newly identified microorganism and wants to investigate whether similar proteins exist in other species.

Which biological database(s) would you recommend, and what information would you expect to obtain from each database? Explain your reasoning.

If a biotechnology company has isolated a protein from a newly discovered microorganism, I would recommend using the [UniProt database](#) and the [Protein Data Bank \(PDB\)](#) to investigate whether similar proteins exist in other species.

[UniProt](#) is one of the most comprehensive protein databases available. It contains protein sequences, functional descriptions, information about biological roles, and links to scientific literature. By searching the newly isolated protein sequence in UniProt, researchers can identify proteins with similar sequences in other organisms. This can provide clues about the protein's possible function and evolutionary history.

The [Protein Data Bank \(PDB\)](#) is also valuable because it stores three-dimensional structures of proteins and other biological molecules. If a structurally similar protein is found in PDB, researchers can gain insights into how the protein works at the molecular level.

Additionally, using [BLAST](#) against protein databases can help identify homologous proteins in different species. Similar proteins often share similar functions, even when they originate from different organisms.

For example, if the isolated protein closely resembles an enzyme found in several bacterial species, researchers may infer that it performs a similar biochemical role. By combining sequence information from UniProt, structural data from PDB, and similarity searches using BLAST, the company can better understand the protein's function and determine how widespread it is across different species.

Question 4

A scientist has identified an unknown protein sequence and wants to determine its function. Which biological database(s) would you recommend for this purpose? Justify your answer by explaining the type of information each database provides.

When a scientist discovers an unknown protein sequence and wants to determine its function, several biological databases can be used together to obtain reliable results.

The first database I would recommend is UniProt. This database provides detailed information about protein sequences, functions, biological processes, cellular locations, and associated scientific studies. Searching the unknown sequence against UniProt can reveal similar proteins whose functions are already known.

Next, I would use the NCBI BLAST protein search tool. BLAST compares the unknown sequence with millions of known protein sequences and identifies similarities. If the sequence closely matches a characterized protein, the scientist can infer its likely function.

The Protein Data Bank (PDB) can also provide structural information if similar protein structures are available. Since protein structure is closely related to function, structural similarities can offer additional evidence about the protein's role.

For example, if the unknown sequence contains a kinase domain identified through InterPro and matches known kinases in UniProt, it is likely involved in phosphorylation processes. Using multiple databases together increases confidence in functional predictions and provides a more complete understanding of the protein.

Question 5

You have been provided with an unknown biological sequence but are not told whether it is DNA, RNA, or protein.

Describe the step-by-step approach you would follow to identify the sequence using biological databases. Explain which databases you would use at each stage and why.

If I were given an unknown biological sequence without knowing whether it was DNA, RNA, or protein, I would follow a systematic approach using biological databases and sequence analysis tools.

First, I would examine the sequence composition. If it contains only the nucleotides A, T, G, and C, it is likely DNA. If it contains U instead of T, it is probably RNA. If the sequence includes amino acid symbols such as M, L, K, or F, it is likely a protein sequence.

Next, I would perform a BLAST search. If I suspect the sequence is DNA or RNA, I would use BLASTN against nucleotide databases such as GenBank. If I suspect it is a protein, I would use BLASTP against protein databases like UniProt. BLAST can identify similar sequences and provide clues about the sequence type and biological origin.

After finding related sequences, I would consult *GenBank* for nucleotide information or *UniProt* for protein-related information. These databases provide annotations, gene descriptions, and biological functions.

If the sequence appears to encode a protein, I would analyze it using InterPro to identify conserved domains and functional motifs. I might also check the *Protein Data Bank (PDB)* for related protein structures.

Finally, I would compare all collected evidence to determine the sequence type, source organism, and possible function. This step-by-step approach ensures accurate identification and allows researchers to make informed biological interpretations even when starting with very limited information.